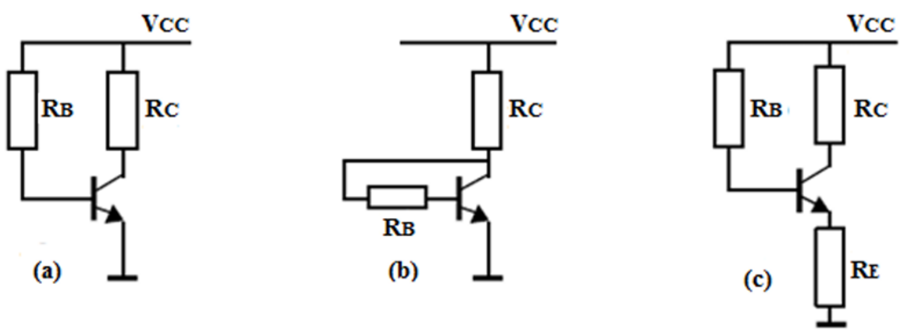
Saad Dahlab University of Blida
Computer Science Department -1st Year LMD-
Semester : 2

Academic year2025/2026
Subject: Fundamental Electronic

Exercise Series 3
Bipolar Transistor

Exercise 1 (course)

Consider the transistor biasing circuits shown below. The given data are: V_{CC}=15V, V_{BE}=0.7 V, Rc=1KΩ, R_B=220 KΩ , R_E=110 Ω.

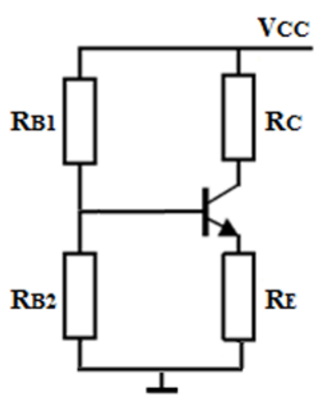


1. For each of the three configurations, calculate the current $I_C$for $\beta = 100$and then for $\beta = 300$.
2. Which configuration is the least sensitive to variations in β?

Exercise 2

The figure below represents a **biasing circuit of an NPN transistor** using two base resistors.
Given:
$\beta = 15$, $V_{BE} = 0.7\ V$, $R_1 = 6.8\ k\Omega$, $R_2 = 2.2\ k\Omega$, $R_C = 3\ k\Omega$, and $R_E = 2\ k\Omega$.

1. Transform the biasing circuit seen from the **base of the transistor** into its **Thevenin equivalent**, and calculate $V_{TH}$and R_{TH}.
2. Calculate the collector current $I_C$for $\beta = 100$and $\beta = 300$. Draw a conclusion.



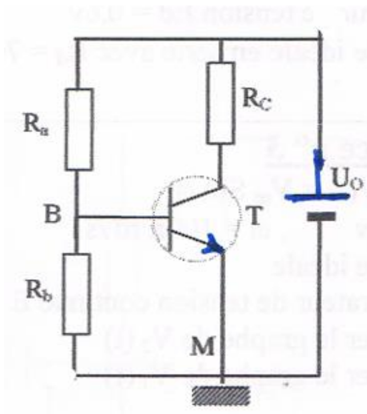
Exercise 3

Using Kirchhoff's law (node law), find the expression relating:
V_{BE}, I_B, U_0, R_a,and R_B.

2/ Find the same expression using **Thevenin's theorem** applied between points **B and M**.

3/ Determine the operating point (Q point)
(at the middle of the DC load line)

given: $U_0 = 10\,V$, $V_{BE} = 0.6\,V$, $I_B = 35\,\mu A$, $R_C = 1\,k\Omega$, $R_a = 200\,k\Omega$, $R_b = 50\,k\Omega$.



Exercise 4 (course)

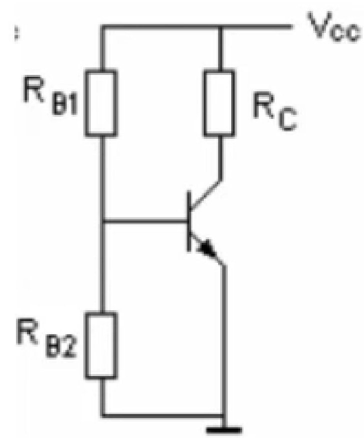
A silicon **NPN transistor** is biased using a **voltage-divider base bias circuit**, as shown in the figure opposite.

Given: **β = 100** and **VCC = 10 V**.

The desired operating point is characterized by:

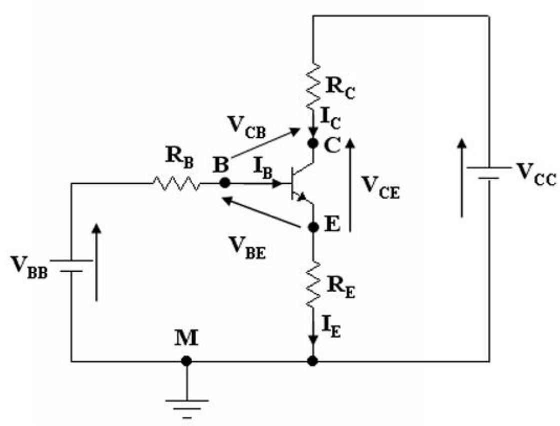
- **V_{CE0} = 5 V** , **I_{C0} = 1 mA** and **V_{BE0} = 0.7 V**

1. Determine the **Thevenin equivalent circuit** between the transistor base and ground.
2. Deduce the equation of the **static input line**.
3. If **R_{B1}** must be ten times greater than **R_{B2}**, calculate the values of **R_{B1}** and **R_{B2}**.
4. Derive the equation of the **static load line**.
5. Calculate the value of the collector resistor **R_C**.



Exercise 5

Consider the circuit shown in the following figure:



1. Calculate the operating point values (**I_B, I_C, and V_{CE}**).
2. Derive and plot the equations of the lines: **$I_C = f(V_{CE})$** and **$I_B = f(V_{BE})$**.
3. Represent the operating point on its corresponding lines.

Given data:

$\beta = 180$, $V_{BB} = 5$ V, $V_{CC} = 10$ V, $V_{BE} = 0.6$ V, $R_B = 10$ kΩ, and $R_C = R_E = 100$ Ω.